



Society of Petroleum Engineers

SPE-228945-MS

Digital 4D Modelling for De-Risking High-Integrity-Risk Wells: A Case Study

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The effective risk assessment of wells with high integrity risk is a crucial step for safe operations in workovers and decommissioning. Traditional methodologies, such as qualitative risk matrices or BowTie analyses, do not provide robust and reliable assessments due to their subjective nature and susceptibility to misinterpretation and bias. This paper illustrates how a digital 4D modelling methodology (i.e. integrated 3D well P&A and subsurface models over time) provides a superior, data-driven approach, that delivers quantitative insights to support operational and regulatory decision-making.

The study presents application of an innovative modelling approach for well integrity assessment in high-risk wells, integrating numerical simulation to enhance the accuracy and reliability of risk evaluations. The proposed model quantifies well-specific leakage risks to surface and crossflow under various reservoir recharge and leaking fluid scenarios. By incorporating probabilistic assessments and accounting for uncertainties such as cement flow properties and casing corrosion, the workflow enables comprehensive risk evaluation and comparison. The detailed and auditable results support effective stakeholder communication and informed decision-making.

The application of the modelling approach is demonstrated using a case study featuring a high-risk well situated in a depleted reservoir with multiple flow zones. A probabilistic analysis, while accounting for uncertainties in cement flow properties, reveals the prevalence of crossflow risks into overburden flow zones while the risk of leakage to surface is expected to be small. The model is also able to quantify the impact of various uncertainties such as casing corrosion, as well as formation porosity, permeability, net-to-gross and reservoir recharge pressure on the subsequent risk of crossflow.

By providing a more robust, data-driven risk assessment, the simulation enables effective comparison of alternative scenarios to ensure compliance with local regulations and company standards. Additionally, it enhances transparency in decision-making by clearly communicating assumptions and findings to regulators and other stakeholders. Ultimately, this approach facilitates the selection of the optimal "As Low As Reasonably Practicable" (ALARP) scenario for remedial actions, alternative P&A designs, monitoring strategies and project planning.

In conclusion, the presented workflow offers significant advancement in well integrity assessment, equipping stakeholders with a powerful tool for informed decision-making, regulatory compliance, and effective risk mitigation in well operations.

Introduction

Well integrity assessment is critical in the oil and gas industry to ensure the safe and efficient operation of wells throughout their lifecycle. It involves evaluating the structural and operational condition of well components, such as casing, cement, and tubing, to prevent leaks, blowouts, or environmental contamination. Regular assessments help identify potential failures, such as corrosion or mechanical damage, before they escalate into costly or hazardous incidents. By maintaining well integrity, operators can protect personnel, preserve environmental safety, comply with regulatory standards, and optimize production, ultimately extending the well's operational life and minimizing financial risks.

This study illustrates the application of a recently developed modelling framework to quantify risk of crossflow and leakage under various scenarios. The framework is applied to assess risk of hydrocarbon leakage from a candidate P&Ad well. The well is located in the North Sea, offshore Netherlands. Assurance of the integrity of the well is critical as the well is located in a fault block, that may communicate with adjacent blocks, posing a risk of recharge over time. Although wells located in other blocks have fully compliant P&A, there is still concern about the quality and longevity of the P&A in this target well.

The well consists of a main bore and a deviated side track well as shown in Figure. The side track has a compliant P&A but the main bore lacks a compliant P&A.

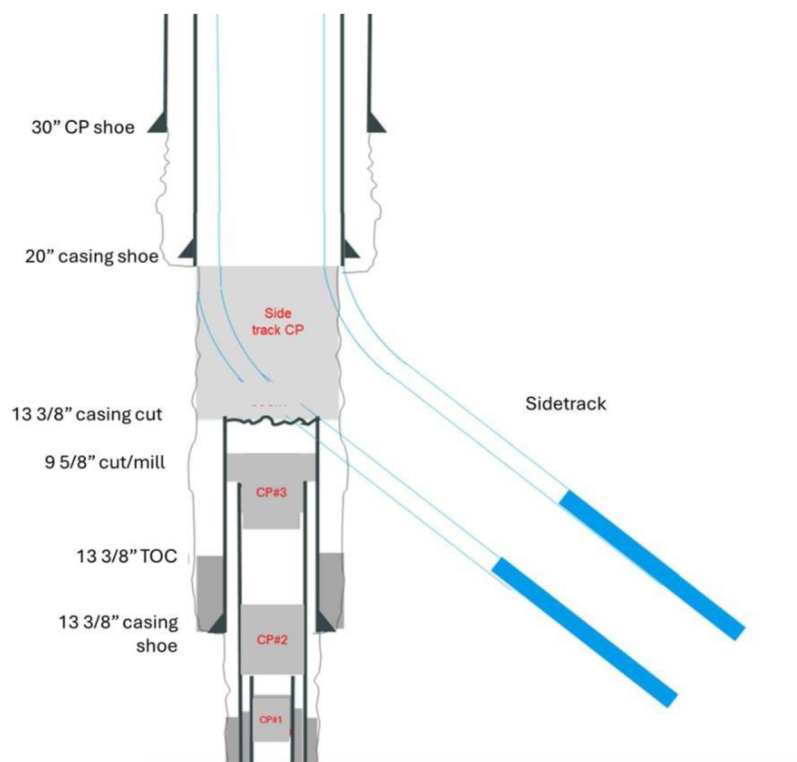


Figure 1—Schematic of the studied well along with the plugs used to abandon the well. The side track part of the well is shown.

The objective of this study is thus, to quantify and compare risk of leakage and crossflow (magnitude and probability) for this well under various reservoir pressurisation scenarios.

The rest of this paper is organised as follows: first, a review of well P&A modelling framework used to quantify risk of leakage in this well is presented. Next the target well is introduced along with assumptions used to model leakage from this well. Finally, the results of the modelling under various scenarios will be presented.

Review of Well P&A Modelling Framework

We use a numerical well P&A model to assess the risk of gas leakage to surface as well as fluids crossflow between formations. The well P&A model essentially translates all well flow paths into a numerical, spatially discretised, grid-based, digital representation (Figure 2). Multiple grid cell types represent the various well components (e.g. intact and impaired cement, casing, tubing, reservoir formation and fluid-filled annular space). Appropriate fluid flow properties are then allocated to each cell. The model solves the cell-to-cell flow from source-to-sink (reservoir-to-surface or crossflow to other zones) for the entire well P&A system over long time periods (typically 3,000 years). Short- and mid-term transient, unsteady flows as well as the long-term, steady state flow rates are then calculated using a finite difference flow simulator as the back engine. More details about the well P&A modelling framework are available in [Jonhson *et al* \(2021a\)](#) and [Jonhson *et al.* \(2021b\)](#)

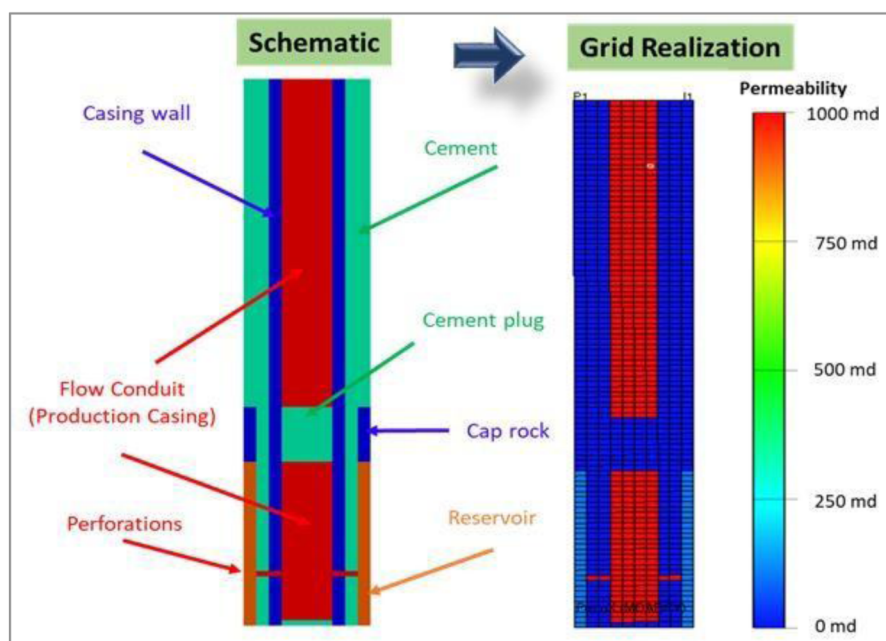


Figure 2—Left: The P&A well schematic and Right: the discretised, numerical flow model ([Johnson *et al*, 2021a&b](#))

The modelling framework has been successfully field tested in several risk-based well P&A designs and integrity assessments (e.g. [Sefat *et al*, 2024](#), [Fraser & Sefat, 2022](#)). A recent upgrade made the framework applicable for CO₂ storage projects where various CO₂ associated physics, such as corrosion, geomechanics and cement geochemistry, may affect the integrity of the P&A wells ([Ghanbari *et al*, 2024](#)).

Target Well Configuration and Conversion into Digitized Version

As a first step, the well schematic diagrams and other relevant information extracted from numerous well reports and reservoir studies were converted into a digitised version suitable for numerical modelling (Figure 33). In generating the digitised version, it is assumed that the lower completion is fully degraded over time and thus, is not explicitly modelled.

The digitised versions include well model with all components and barriers as well as near well and reservoir sub-models for the connected formations with flow potential. Intervals with good and poor cement, well casings, formations with flow potential, etc. are marked accordingly.

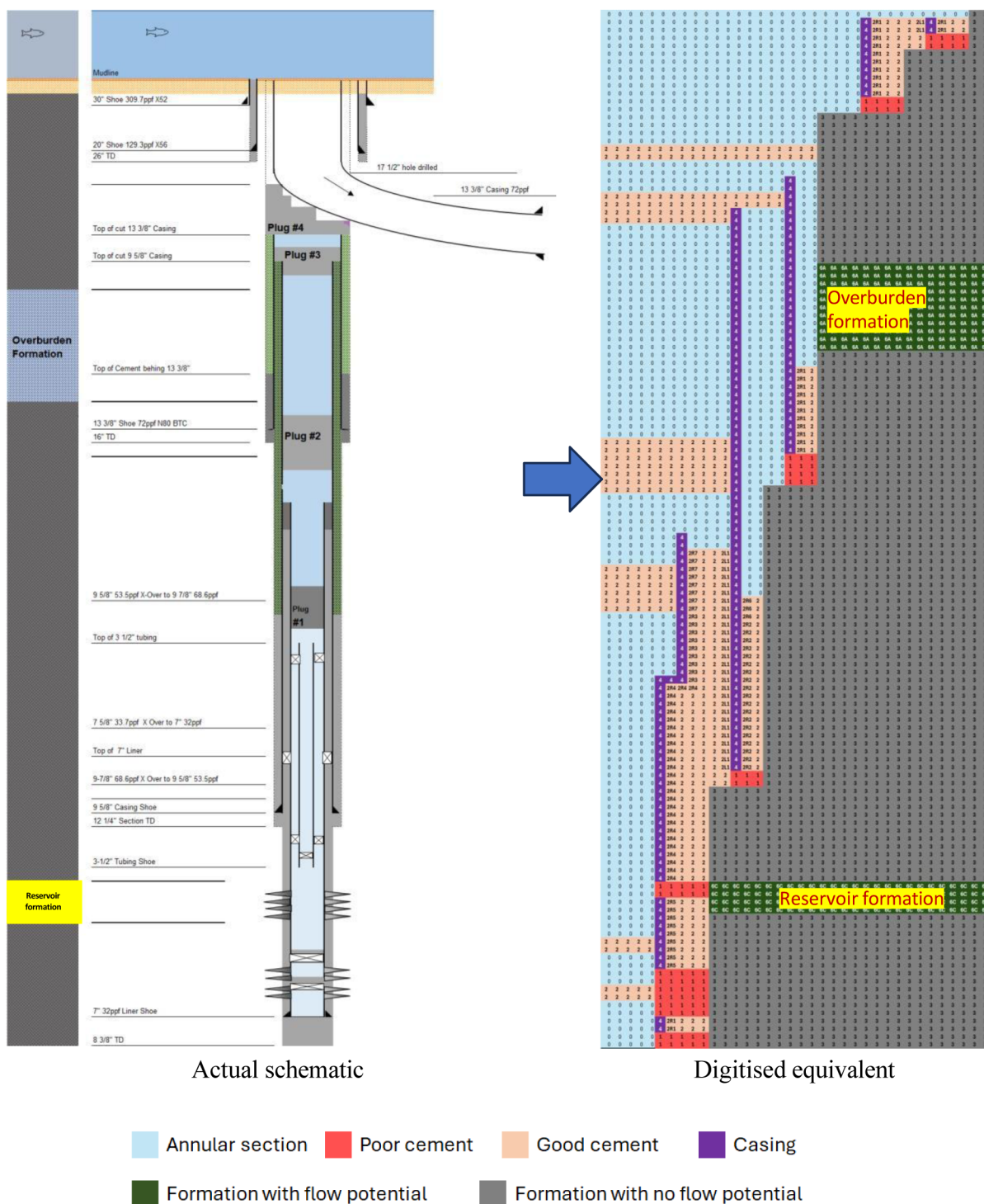


Figure 3—Left: Actual configuration of the target well. Right: Equivalent digitised version. Cells in the digitised version will be allocated with appropriate flow properties. The digitised version is not to scale.

The target well is connected to two permeable formations: 1) a deeper reservoir formation full of gas (methane) and 2) a shallower overburden formation full of water. Other formations in the well are not explicitly modelled as they are considered to have no flow potential.

Appropriate flow properties (porosity and permeability) were then allocated to each numerical element. For example, the wellbore filled with sea water (shown in blue) is represented by the sea water's density value and a very high permeability and porosity. It thus, provides a minimal resistance to flow. Formation properties were also provided by the field's reservoir engineers.

Characterisation of flow properties for cement and defects are particularly important as they constitute major leakage path for the reservoir fluid. Intact cement permeability is assumed as 0.002mD though its effect is expected to be limited compared to the flow properties of cement defects.

Cement defects on the other hand have major impact on the degree of leakage and crossflow and they should be carefully characterised. In this study cement defects are represented as equivalent microannuli (MA) between cement and adjacent tubulars. The size of MA can be deduced from cement logs or cement reports. Cement logs can be used to categorise the quality of the bond between cement and its adjacent casing into 4 categories of very good, good, medium and poor bond. For each bond category, MA is expected to vary within a particular min/max range (Ford *et al*, 2022) as shown in Table 1.

Table 1—Allocated MA sizes based on the Equivalent Cement Quality based on cement logs (after Ford *et al*, 2022).

Colour	Equivalent Cement Quality	Acoustic Impedance (AI)_Min	Acoustic Impedance (AI)_Max	MA_min (μm)	MA_max (μm)
Black	Very good	5	6	0.0	0.1
Dark Brown	Good	3.85	5	0.1	3.2
Light Brown	Medium	2.7	3.85	3.2	82.2
Yellow	Poor	2.3	2.7	82.2	255.0

In this well, cement logs have been available for behind the 7" and 9 5/8" casing strings and have been used to characterise the respective categories of cement sheath bond qualities along those casing strings (Figure 4). For each section, the available cement logs have been used to first categorise them into very good/good/medium/poor cement bond quality sections. For each category, a random microannuli value was then generated within the range respective for that category (as shown in Table 1). Figure 4 shows a generated sample microannuli size for this well. Different colours (green, yellow, red) correspond to different cement bond quality categories. Using this approach, the variability in the cement bond quality can be best captured toward modelling.

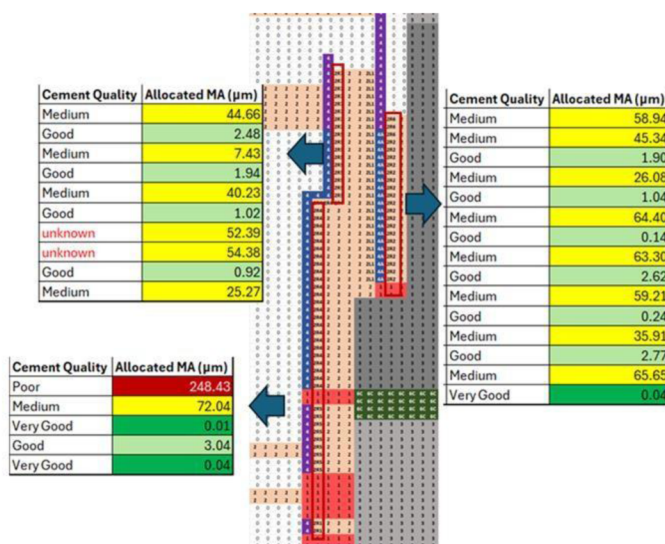


Figure 4—Random defect size allocation for each section of CBL reading based on the inferred bond categories identified from CBL. Unknown assumed as medium bond.

For sections where no CBL information was available, a random microannuli between 0-100 μm was assumed. It is also assumed that no microannuli exists between cement plugs and adjacent casing as the plugs have been set on a base and pressure tested.

Since the well has been in prolonged contact with reservoir fluid (which may contain traces of CO_2 and other corrosive fluids), a conservative assumption has been made that part of the casing without a full lateral barrier along 7", 9 5/8" and 13 3/8" casing strings will be corroded and thus, flow could be possible across those corroded casing section. It is also conservatively assumed that corrosion degrades the respective casing sections from the very beginning of modelling. The assumed corroded sections of casings are indicated in red colour in Figure 5.

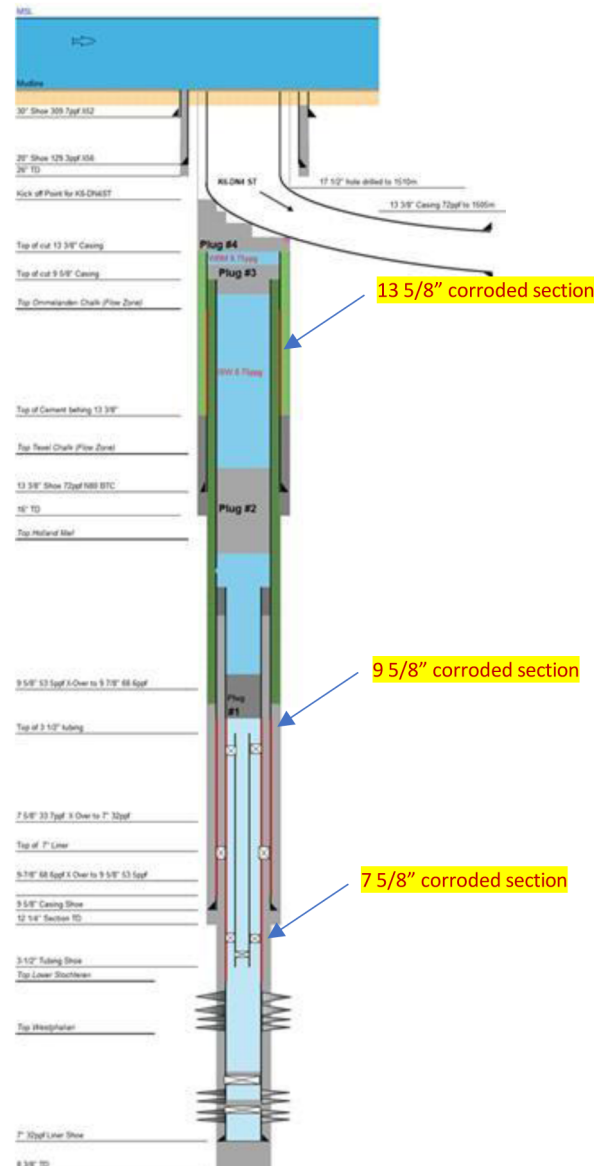


Figure 5—Schematic of the target studied well where likely corroded sections of casing by reservoir fluids are highlighted in red.

A few further important assumptions were made for this modelling.

- It is assumed that any weighted mud, will eventually settle, thus the seawater density was assumed in the annular regions.

- Modelling is also assumed to be isothermal at reservoir temperature.
- The reservoir formation is assumed charged conservatively with methane only (i.e. no water assumed existing in the reservoir formation) to illustrate the extreme gas mobility condition.
- A fracture pressure gradient of equivalent to 1.4sg was also used in this modelling. That indicates at shallow depths, there is possibility of formation failure as wellbore fluid pressure may exceed the fracture pressure at sufficiently low depths. This could occur due to gas or water accumulation underneath the shallowest plug.

Once the digitized version is constructed, it is then attached to a finite difference, flow simulator. This study's key outputs to support decision making are (1) gas/water leakage to surface, (2) inter-layer crossflow within the well and (3) the pressure profile along the well.

Three P&A scenarios have been studied:

- A scenario where reservoir pressure is assumed at the initial virgin pressure at the start of simulation and remained constant for the entire duration of modelling (both deterministic and probabilistic approaches).
- A scenario where reservoir pressure evolution, as provided by subsurface reservoir simulation, is assumed (only deterministic).
- A further sensitivity analysis to identify the threshold reservoir pressure after which methane leakage may occur.

Results

Methane Leakage under Assumed Virgin Reservoir Recharge Pressure

The virgin reservoir pressure before any depletion occurring has been reported as 375bar. As a first step modelling, gas and water leakage and crossflow were studied under assumed virgin reservoir pressure of 375bar. For this analysis, it was assumed that corrosion has fully degraded the section of casing without a full lateral barrier as shown in [Figure 5](#). That provides understanding of the maximum (worst-case) leakage and crossflow scenario.

[Figure 6](#) illustrates the results. The results include gas crossflow from reservoir formation (top-left panel) into the wellbore and similar crossflow from wellbore into the Overburden (mid-left panel). A fraction of gas also arrives to surface, and this has been reported as well (bottom-left panel). Same as gas information, similar parameters have been reported for water as well (right panels).

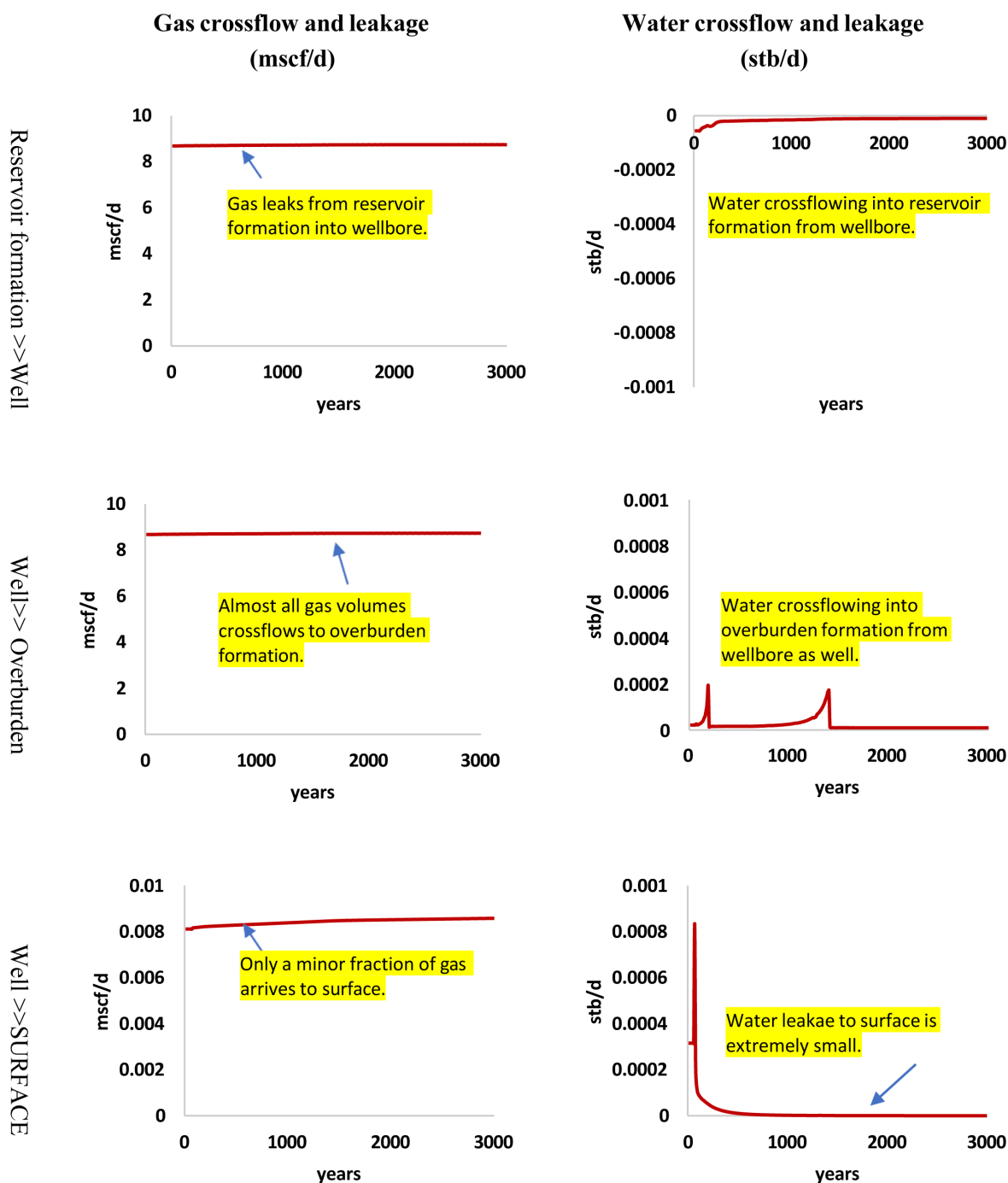


Figure 6—Gas (left column) and water (right column) leakage and crossflow profiles under assumed fixed reservoir formation pressure (375bar).

The crossflow and leakage profiles shown in Figure 6 confirm that under assumed fixed pressure of 375bar and corrosion occurring in all three casing strings (as shown in Figure 5), the risk of methane leakage to surface minimally exist (around 8 scf/day). Results indicate the main risk is expected to be gas crossflow between reservoir formation and overburden. Comparing gas crossflow and leakage to surface values, it can be seen that only a very minor fraction of leaked gas from reservoir formation arrives to surface and the rest crossflows to the overburden. Gas crossflow between the reservoir and the overburden is more likely than leakage to the surface, as in this well lower pressure is required for gas to migrate through overburden formation with flow potential, which might also be fractured.

Figure 7 shows major leakage path of gas from reservoir formation, into overburden and then to surface.

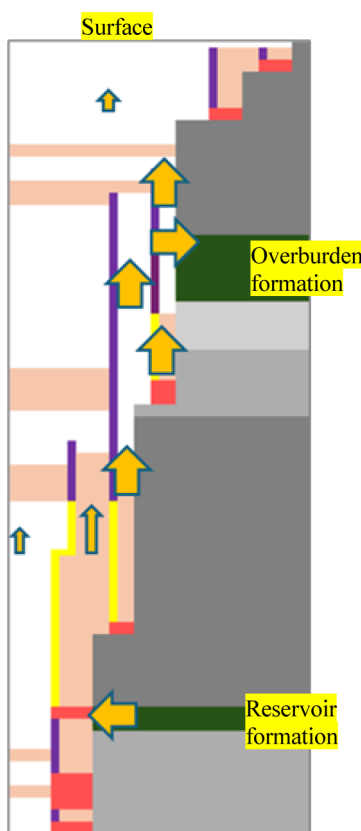


Figure 7—Arrows show major gas leakage path from reservoir formation to surface when corrosion is considered in the well. The majority of leaked gas from reservoir formation crossflows to overburden. Only a small fraction of leaked gas arrives to surface. Sections of corroded casings are shown in yellow.

The magnitude of water crossflowing and leakage is expected to be very small in this scenario. Significant gas crossflowing between reservoir and overburden, practically blocks the path of water crossflowing from overburden to reservoir formation.

This initial analysis provides insight into the likely gas crossflow and leakage characteristics between various formations and well and ultimately to surface.

Expansion to Probabilistic Results

Although informative, the above analysis illustrates the performance of the well P&A system under a single realisation scenario. Importantly, cement flow properties are the main uncertainty which will be different from those currently assumed. Hence, a full probabilistic simulation was conducted to understand the leakage and crossflow risk at a broader spectrum of influencing input parameters.

We used Monte Carlo approach to conduct the probabilistic simulation. The uncertain parameters considered include: intact cement permeability (randomly varying between 0.0035 and 0.03mD) and microannuli size, where various realisations based on the expected cement quality, as described in the previous section, were generated. All other parameters were kept fixed as the previous section. A total of 100 realisations were generated and modelled. The monitoring parameters after completing the probabilistic simulations include cumulative gas and water crossflow between well and formations (denoted by reservoir formation>>well and well>>Overburden) as well as water/gas leaking to surface (denoted by Well>>SURFACE). All cumulative parameters were measured after 3000 years.

Figure 8 shows the final probabilistic results. Horizontal axes in the shown panels indicates the cumulative measured values against their respective Cumulative Distribution Function (CDF) values shown

along the vertical axes. For this analysis, in addition to a scenario where all 3 above noted casing corroded, a further scenario where corrosion has not been occurred was also modelled (dashed profiles).

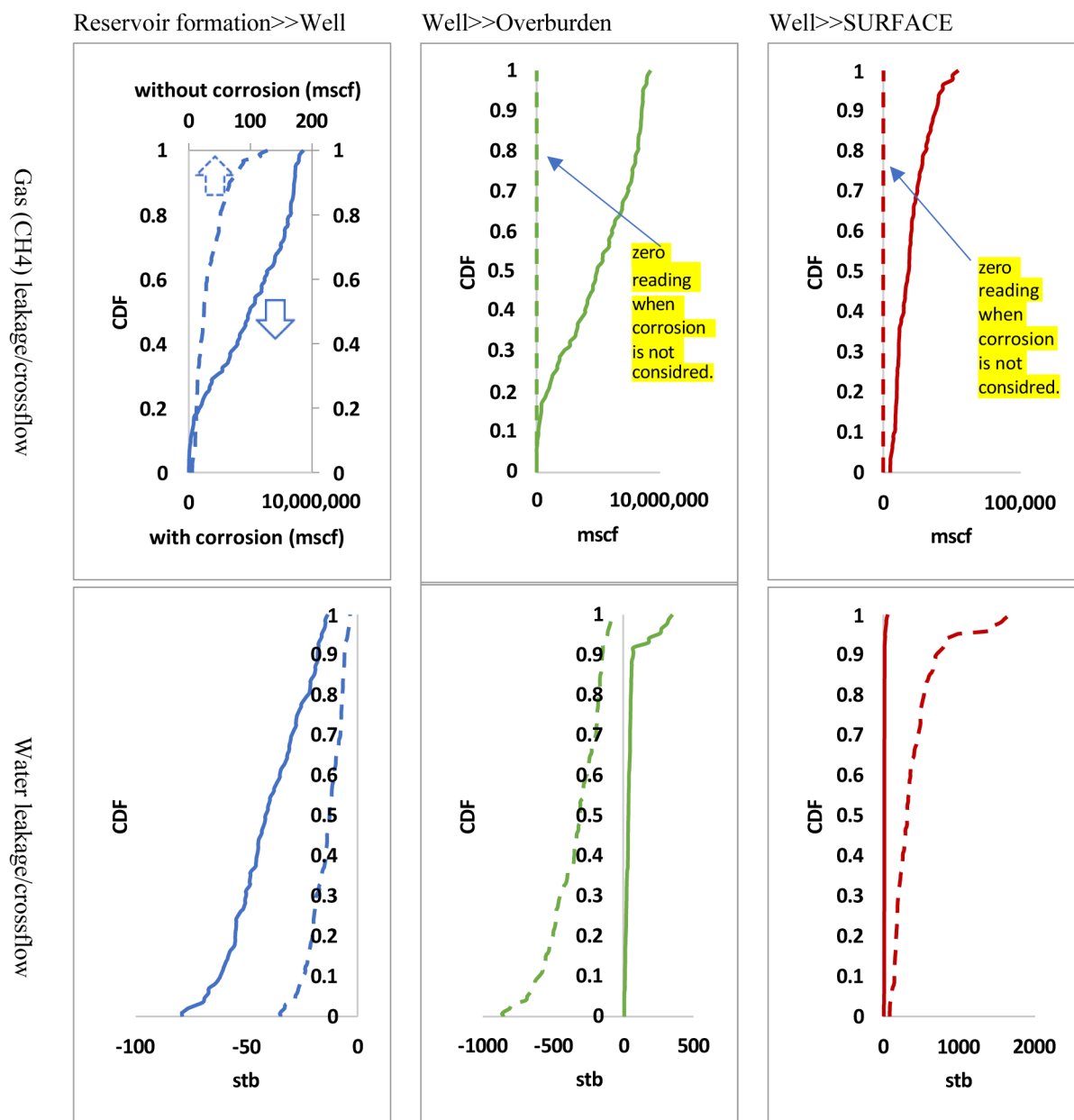


Figure 8—CDF summary of probabilistic results indicating gas/water crossflow and leakage to surface profiles. Dashed and solid profiles illustrate scenarios without and with corrosion, respectively.

Fully corroded scenario (solid profiles): As with the initial deterministic results, the probabilistic results for a corroded casing scenario (shown in Figure 5) confirm that the majority of leaked methane from the reservoir formation is expected to crossflow into overburden and only a small fraction might leak to surface. The risk of water crossflow and leakage is also very small in this case.

No corrosion occurring scenario (dashed profiles): Results show in the absence of corrosion, small volume of gas leaks from reservoir formation into wellbore, however, no gas crossflows into overburden or leaks to surface. Looking at water profiles, there is almost identical reading between the volume of water flowing from overburden into well and water leaking at surface; only a very minor fraction of water crossflows from well into reservoir formation as gas is leaked to the wellbore.

Comparing these results indicates that corrosion could significantly change the major leakage and crossflow path for both gas and water phases. In the absence of corrosion, gas leaks through the main bore of the well while water flows from behind the cement sheath. When corrosion is occurring, gas from reservoir migrates to overburden formation and blocks water crossflow out of overburden. Table 2 shows the P50 summary for various profiles of the two P&A scenarios illustrated in Figure 8.

Table 2—Summary P50's for the scenarios illustrated in Figure 8

	Gas CH ₄ (cumulative Mscf after 3,000 yrs)			Water (Cumulative STB after 3000 years)		
	Reservoir formation >>Well	>>Overburden Well	>>SURFACE Well	Reservoir formation >>Well	>>Overburden Well	>>SURFACE Well
No corrosion considered	24	0	0	-13	-307	316
Corrosion in 3 casing strings considered	5,171,381	5,147,103	20,058	-44	36	15

Leakage under Simulated Reservoir Recharge Pressure Profile

In all the previous results, it was assumed that the reservoir formation pressure is initiated at 375bar as the virgin pressure of the formation. In reality however, based on reservoir simulation studies, it is expected that pressure should have an evolving profile starting from current depleted value up to about 300bar.

A further sensitivity study was thus, conducted to understand the risk of gas and water leakage and crossflow in the studied well under a more realistic near wellbore pressure predicted by field scale reservoir modelling. Figure 9 shows the provided profiles obtained from field scale reservoir modelling. The profile indicates pressure building up from a depleted field pressure of ~101bar to about 300 bar during the 3000 years simulation period.

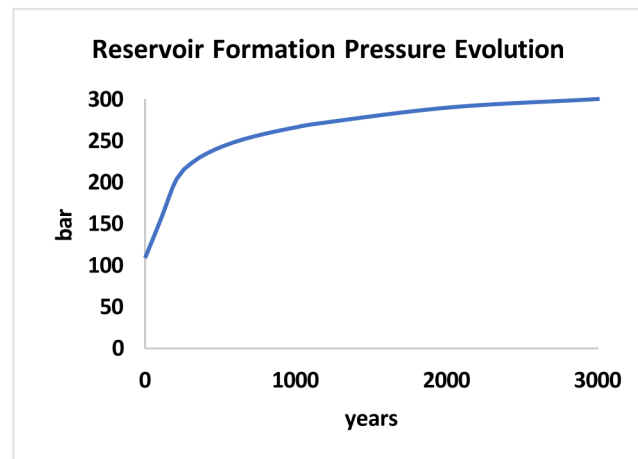


Figure 9—Simulated reservoir pressure profile near the target well.

The well P&A model was upgraded to estimate the risk of leakage and crossflow for this scenario of evolving reservoir pressure. For this modelling, reservoir inflow pressure was updated periodically every 50 years over 3000 years modelling period.

The risk of leakage and crossflow was then assessed under simulated pressure profiles (Figure 10). The risk has been tested for the scenario when corrosion is occurring in the P&A well system (as shown in Figure 5).

Results now indicate the sub-hydrostatic recharge pressure in reservoir formation, significantly reduces the risk of gas crossflow and leakage to surface. Results indicate during the early period, the reservoir pressure is significantly lower, which helps to alleviate the leakage risk. There is no gas leakage from reservoir formation into well and consequently no gas crossflows into overburden or leaks to surface. In terms of water, results indicate a significant water crossflow between overburden and reservoir formation. Comparing with those results presented in previous section (Figure 8 with Figure 10), water crossflow increases since pressure in reservoir formation is now significantly reduced compared to the original assumption of 375bar fixed.

The current results indicate that recharge pressure is the key parameter in this well controlling the responses of the system and risk of leakage and crossflow. With the supplied pressure profile, no risk of gas leakage and crossflow is expected. Also, all profiles are dominated with water crossflow from high pressure overburden formation to low pressure reservoir formation.

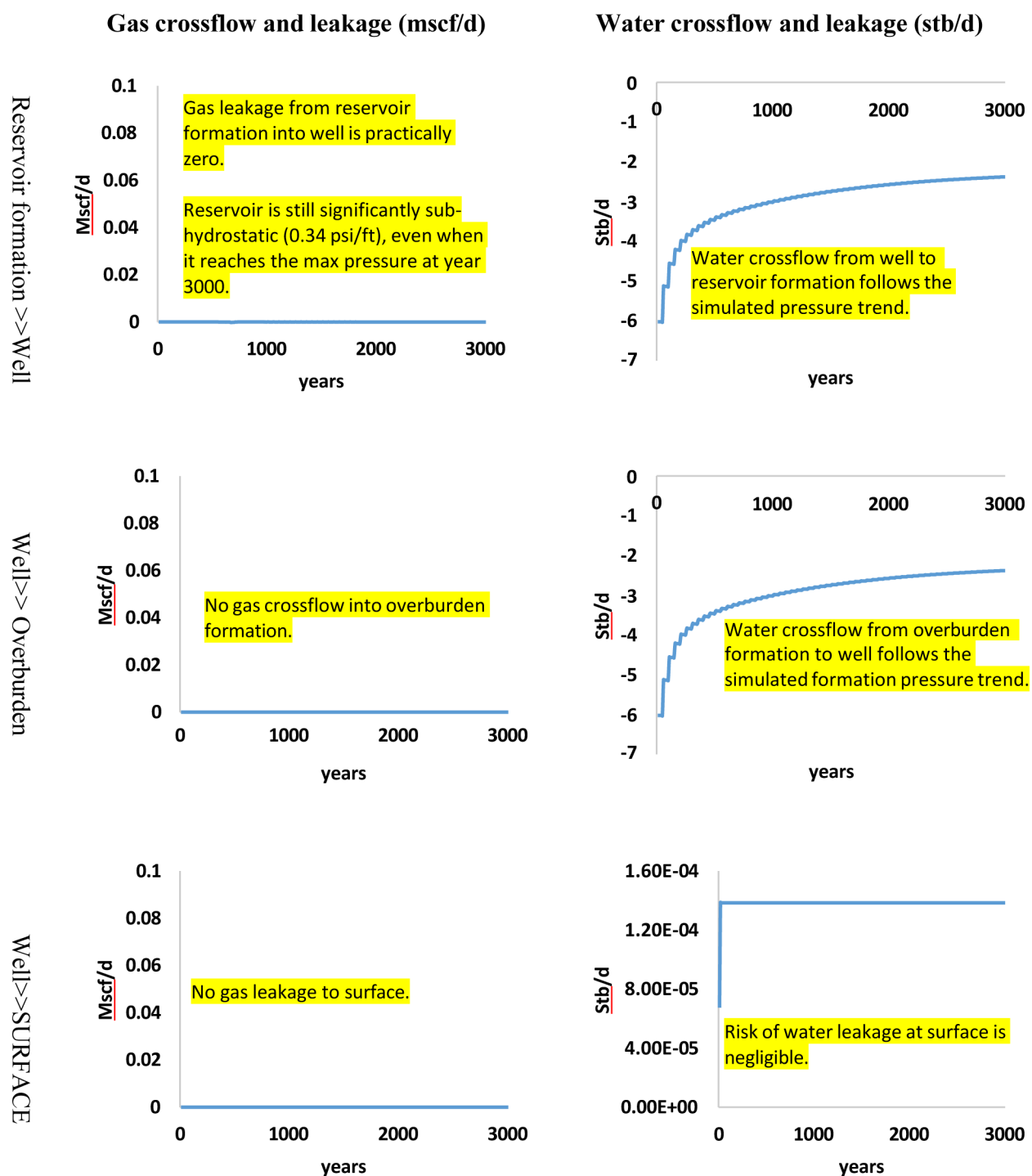


Figure 10—Gas (left column) and water (right column) leakage and crossflow profiles under simulated reservoir formation pressure profile illustrated in Figure 9.

Understanding the Onset Threshold of Reservoir Formation Pressure Allowing Gas Leakage to Surface

The above results illustrate that with the current simulated pressure profile (where pressure remains below hydrostatic pressure) it is unlikely for any gas to leak to surface (Figure 10). However, the previous results - where reservoir formation was charged with methane at fixed 375 bar at the start of simulation shows that methane can possibly leak to surface (Figure 8).

It will thus, be useful if the threshold pressure between 300-375 bar after which leakage to surface occurs to be identified.

A further sensitivity analysis was then conducted. For this analysis, the reservoir formation pressure assumed initially following the simulated pressure profile up to 3000 years (i.e. max 300 bar by 3000 years), however, it is increased in 5 bar increments up to 375 bar beyond 3000 years to understand the onset thresholds of leakage pressure. A scenario with corrosion was considered for this sensitivity analysis. [Figure 11](#)-left illustrates the assumed reservoir formation pressure and [Figure 11](#)-right shows the observed methane leakage from reservoir formation to wellbore over time.

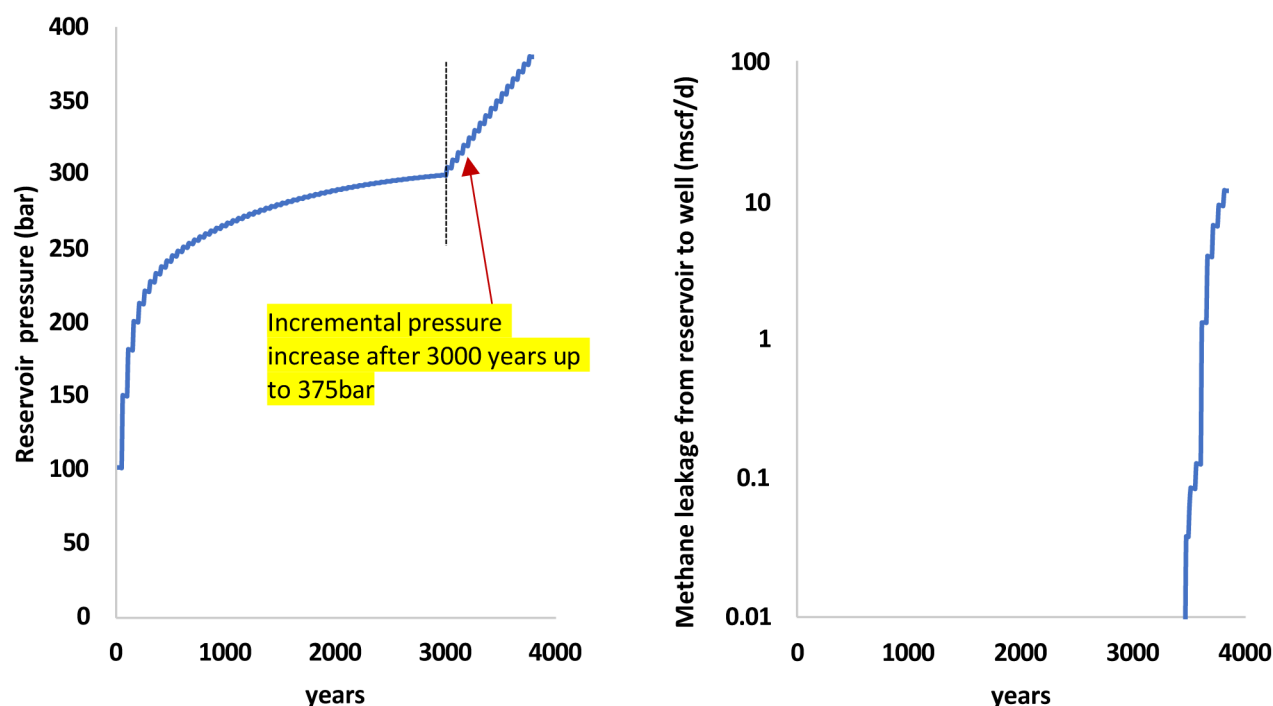


Figure 11—Reservoir formation pressure (left) and corresponding leakage from reservoir formation to wellbore (right). Pressure is increased in +5bar increments to identify the onset threshold of gas leakage occurring at surface.

[Figure 3](#) [Figure 12](#) shows the cross plot of methane leakage rate from formation to wellbore vs. formation pressure, indicating that for the chosen scenario the onset pressure threshold where initial leakage from formation to wellbore happens is about 345bar. That means as long as the formation pressure is expected to remain below 345bar, there is minor risk of gas leaking at surface.

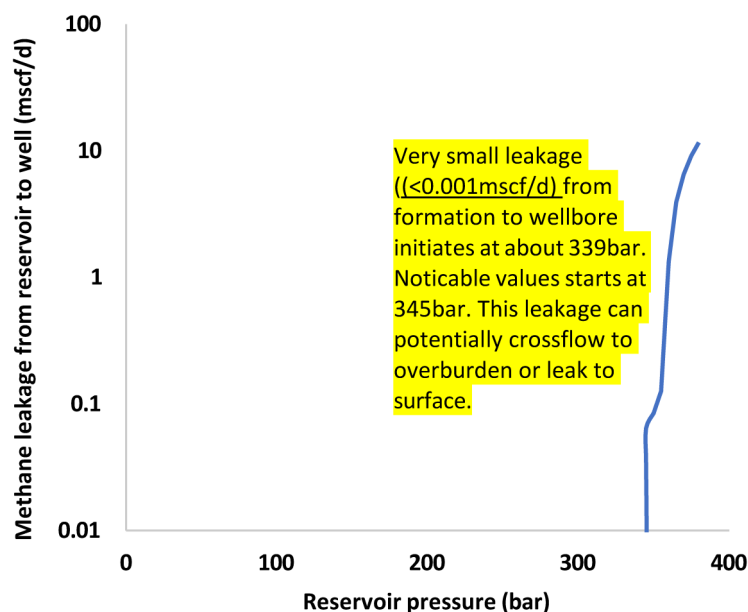


Figure 12—Cross plot of observed leakage at surface versus formation pressure. The onset leakage pressure is estimated at about 345bar.

Conclusions

A numerical well P&A modelling methodology is applied to assess the risk of methane leakage and crossflow in an offshore well. Results indicate that while there is a risk of gas leakage to surface when reservoir is conservatively assumed at virgin reservoir pressure at the start of simulation, under a more realistic pressure profiles simulated by field scale modelling, there is no such risk. A further sensitivity analysis indicated that as long as pressure in the reservoir formation is below 345bar, no risk of gas leakage to surface is expected.

The value of the well P&A modelling framework lies in its ability to quantify risk of factors influencing the well integrity. It thus, offer superior value against more qualitative well P&A risk assessment techniques. The quantitative risk assessment provides significant value to both operators and regulators in terms of understanding the risk of well integrity to facilitate the selection of the ALARP scenario for remedial operations, P&A designs, monitoring strategies and project planning.

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